

Yiqiang Wen

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EDUCATION

China Agricultural University

Beijing, China

Bachelor of Agronomy in Aquaculture

Sept. 2021– July 2025

Overall GPA: 3.65/4.0

Core Curriculum: Biochemistry

Cell & Molecular Biology

Genetics

Histology

Embryology

Microbiology

Zoology

Physiology

Organic chemistry

Institute of Botany, Chinese Academy of Sciences (IB-CAS)

July 2022 – Dec. 2023

State Key Laboratory of Systematic and Evolutionary Botany

Joint Research Training Program, Supervisor: Wenli Chen (Associate Researcher)

RESEARCH EXPERIENCE

I. Protein Biochemistry and Disease Mechanisms

Research Assistant, China Agricultural University, Supervisor: Dong Yang (Associate Professor)

July 2025 – Present

Investigation of the Relationship Between Redox State and Protein Stability in Multi-Domain Proteins

Collaborative project with computational protein modeling groups at the University of Illinois at Urbana-Champaign

- Investigated the effects of redox-dependent disulfide bond formation on protein conformation and stability through comparative analysis of diverse protein systems.
- Contributed to plasmid design and expression vector optimization through molecular cloning, plasmid editing, and vector replacement for bacterial expression of 12 single-subunit, multi-domain proteins.
- **Recombinant Protein Expression & Purification:** Optimized expression conditions for multiple proteins to address poor expression and inclusion body formation, followed by chromatographic purification to obtain highly purified protein samples for biophysical characterization.
- Characterized protein conformation and stability by comparing secondary structure and thermal stability under oxidizing and reducing conditions using circular dichroism (CD) spectroscopy.

Flavonoids as Multi-Target Modulators in Alzheimer's Disease

- Investigated the therapeutic potential of flavonoids in Alzheimer's disease, focusing on their effects on multiple pathological processes, particularly their ability to disaggregate pre-formed A β fibrils.
- Screened 17 flavonoids via ThT fluorescence assays and confocal microscopy to expand evidence for flavonoid-mediated A β fibril disaggregation, identifying lead compounds.
- Conducted systematic literature synthesis and structure–activity relationship (SAR) analysis, classifying flavonoid mechanisms of action across multiple AD-related targets and highlighting their therapeutic potential in Alzheimer's disease, **culminating in a first-author review article.**

II. Biological Systems and Molecular Research

Functional Analysis of Plant Cell Surface Receptors (LRR-RLKs)

University of Toronto, Mitacs Globalink Research Internship, Supervisor: Adam Mott

July 2024 – Oct. 2024

- Performed phenotypic screening of BRR (broad-range resistance genes) knockout mutants under pathogen-associated and abiotic stress conditions to investigate receptor-mediated immune responses.
- Designed and conducted stomatal and water-loss assays, analyzing phenotypic data to test drought-response hypotheses and support a potential role of BRRs in drought tolerance.

Phylogenetic and Systematic Research about Needle Grasses (Poaceae: Pooideae: Stipeae)

IB-CAS, Supervisor: Wenli Chen (Associate Researcher)

July 2022 – Feb. 2025

- Sorted and cataloged plant specimens for the National Herbarium (PE), constructed *Poaceae* database.
- Defined the systematic and evolutionary relationships of the genus *Neotrinia* within the needle grass tribe and conducted taxonomic revisions through both fieldwork and laboratory research.

- Integrated multiple taxonomic evidence, including the analysis of phylogeny tree of chloroplast genome, ITS, cytogenetics (chromosome), and lemma micromorphology to resolve phylogenetic relationships within the genus *Neotrinia*.

Eco-Friendly Protein Source: Black Soldier Fly Larvae as a Fishmeal Substitute

China Agricultural University, Supervisor: Songjian Nan (Associate Professor)

Oct. 2022 – Oct. 2024

- Investigated growth characteristics of black soldier fly larvae reared on mixed bran-tofu residue substrates.
- **As project leader**, secured project funding and designed and conducted feeding trials to evaluate growth performance, intestinal enzyme activity, and gut microbiota in crucian carp.
- Analyzed feeding trial data to determine the maximum tolerated and optimal replacement levels of fishmeal with black soldier fly larvae meal in crucian carp diets.

INTERNSHIP

Yantai Institute of Coastal Zone Research, Chinese Academy of Sciences

Aug. 2023 – Dec. 2023

Supervisor: Wenjun Li (Associate Researcher)

- Studied the protein structure and **disease treatment** effectiveness of **phycocyanin** and **phycocyanobilin**.
- Conducted literature review and developed purification protocols for phycocyanin, assisting in protein purification and preparation for downstream in vivo studies.
- Participated in mouse experiments, including oral gavage, physiological measurements, and sample collection to investigate the therapeutic potential of phycocyanin in pulmonary fibrosis-related models.

PUBLICATIONS & PATENTS

Yiqiang Wen, Dong Yang. Flavonoids for the whole course of treatment of Alzheimer's disease. (Manuscript in preparation for submission)

Haoyang Zhang, **Yiqiang Wen**, Junpeng Qu, Songjian Nan. Study on the growth characteristics of the larvae of the Black soldier fly in the mixed substrate of bran and bean curd residue. *Hubei Agricultural Science*. 2024, 63 (7): 129-133, 153.

Yiqiang Wen, Haoyang Zhang, Junyang Li, Songjian Nan.02322532515.3: The invention relates to an intelligent assembled larval breeding device of the *Hermetia illucens*. 2023.9.18. Chinese Patent.

Haoyang Zhang, **Yiqiang Wen**, Junyang Li, Songjian Nan. 202322532458.9: The invention relates to an intelligent welfare physiological monitoring chicken coop. 2023.9.18. Chinese Patent.

CONFERENCES ATTENDED & SUMMER PROGRAMS

The 5 th Sino-German Block Seminar of IRTG AMAIZE-P, <i>University of Hohenheim, Germany</i>	2023
Summer School: Plant Development and Molecular Biology, <i>Peking University, China</i>	2023

HONORS & AWARDS

First Prize, <i>National Undergraduate Competition on Agriculture, Architecture, Environment, and Energy Engineering</i>	2024
Merit Student, <i>China Agricultural University</i>	2024
Literary and Sports Merit Scholarship, <i>China Agricultural University</i>	2024
Third Prize in Biochemistry Laboratory Skills, <i>Biochemistry Innovation & Entrepreneurship Competition</i>	2023
Second Prize (Top 10%) Academic Scholarship, <i>China Agricultural University</i>	2023
Excellent Student Leader, <i>China Agricultural University</i>	2022

TECHNICAL SKILLS

Protein Biochemistry: Recombinant protein expression, protein purification (affinity, ion-exchange, and size-exclusion chromatography), ÄKTA Purifier (FPLC), SDS-PAGE, fluorescence-based aggregation assays (ThT), circular dichroism (CD) spectroscopy, protein quantification (BCA), mass spectrometry verification

Molecular Biology: Molecular cloning, plasmid construction, sequence analysis and verification, PCR primer design, bacterial transformation, agarose gel electrophoresis, genotyping, quantitative PCR (qPCR)

Microscopy & Imaging: Confocal fluorescence microscopy, electron microscopy (SEM/TEM)

Data Analysis & Software: GraphPad Prism, SnapGene, SPSS, ImageJ, Adobe Illustrator, Adobe Photoshop, PyMOL, webpage development, SolidWorks (CAD), Visual Basic